

Review of spectroscopic methods

18

Connections

➡ Building on

- Mass spectrometry **ch3**
- Infrared spectroscopy **ch3**
- ^{13}C NMR **ch3**
- ^1H NMR **ch13**
- Stereochemistry **ch14**
- Conformation **ch16**
- Elimination **ch17**
- Carbonyl chemistry **ch10** and **ch12**

Arriving at

- How spectroscopy explains the reactions of the $\text{C}=\text{O}$ group
- How spectroscopy tells us about the reactivity of, and reaction products from, conjugated $\text{C}=\text{C}$ and $\text{C}=\text{O}$ bonds
- How spectroscopy tells us about the size of rings
- How spectroscopy solves the structure of unknown compounds
- Some guidelines for solving unknown structures

➡ Looking forward to

- A final review of spectroscopy, including what it tells us about the stereochemistry of molecules **ch31**
- Spectroscopy is an essential tool and will be referred to throughout the rest of the book

This is the first of two review chapters on spectroscopic methods taken as a whole. In Chapter 31 we shall tackle the complete identification of organic compounds, including the vital aspect of stereochemistry, introduced in Chapters 14 and 17. In this chapter we gather together some of the ideas introduced in previous chapters on spectroscopy and mechanism, and show how they are related. We shall explain the structure of the chapter as we go along.

There are three reasons for this chapter

1. To review the methods of structure determination we met in Chapters 3 and 13, to extend them a little further, and to consider the relationships between them.
2. To show how these methods may be combined to determine the structure of unknown molecules.
3. To provide useful tables of data for you to use when you are attempting to determine unknown structures.

The main tables of data appear at the end of the chapter (pp. 423–425) so that they are easy to refer to when you are working on problems. You may also wish to look at them, along with the tables in the text, as you work through this chapter.

We shall deal with points 1 and 2 together, looking first at the interplay between the chemistry of the carbonyl group (as discussed in Chapters 10 and 11) and spectroscopy, solving some structural problems, then moving on to discuss, for example, NMR of more

than one element in the same compound, doing some more problems, and so on. We hope that the lessons from each section will help in your overall understanding of structure solving. The first section deals with the assignment of carbonyl compounds to their various classes.

Spectroscopy and carbonyl chemistry

Chapters 10 and 11 completed our systematic survey of carbonyl chemistry, and we can now put together chemistry and spectroscopy on this most important of all functional groups.

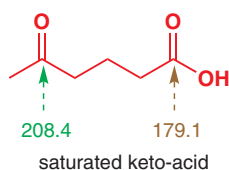
We have divided carbonyl compounds into two main groups:

1. **aldehydes** (RCHO) and **ketones** (R^1COR^2)
2. **acids** (RCO_2H) and their derivatives (in order of reactivity):
 - acid chlorides** (RCOCl)
 - anhydrides** (RCO_2COR)
 - esters** ($\text{R}^1\text{CO}_2\text{R}^2$)
 - amides** (RCONH_2 , R^1CONMe_2 , etc.).

Which spectroscopic methods most reliably distinguish these two groups? Which help us to separate aldehydes from ketones? Which allow us to distinguish the various acid derivatives? Which offer the most reliable evidence on the chemistry of the carbonyl group? These are the questions we tackle in this section.

Distinguishing aldehydes and ketones from acid derivatives

The most consistently reliable method for doing this is ^{13}C NMR. It doesn't much matter whether the compounds are cyclic or unsaturated or have aromatic substituents, they all give carbonyl ^{13}C shifts in about the same regions. There is a selection of examples on the facing page which we now discuss. First, look at the shifts arrowed into the carbonyl group on each structure. All the aldehydes and ketones fall between 191 and 208 ppm regardless of structure, whereas all the acid derivatives (and these are very varied indeed!) fall between 164 and 180 ppm. These two sets do not overlap and the distinction is easily made. Assigning the spectrum of the ketoacid in the margin, for example, is easy.



● ^{13}C NMR distinguishes acid derivatives from aldehydes and ketones

The carbonyl carbons of all aldehydes and ketones resonate at about 200 ppm, while acid derivatives usually resonate at about 175 ppm.

^{13}C NMR shifts of carbonyl groups

Carbonyl group	δ_{C} , ppm
aldehydes	195–205
ketones	195–215
acids	170–185
acid chlorides	165–170
acid anhydrides	165–170
esters	165–175
amides	165–175

aldehydes aromatic aldehyde: vanillin conjugated unsaturated aldehyde: all <i>trans</i> retinal	ketones cyclic conjugated ketone: (-)-carvone saturated ketone: raspberry ketone
acids saturated: lipoic (thioctic) acid conjugated: shikimic acid aromatic: salicylic acid non-conjugated: ibuprofen	
acid chlorides saturated: acetyl chloride conjugated unsaturated: maleic anhydride	anhydrides two signals: 169.5, 177.2 saturated cyclic: maleic anhydride unsaturated conjugated cyclic: maleic anhydride saturated cage tricyclic
esters conjugated: methyl methacrylate ester of aromatic acid: benzocaine	amides simple amide: dimethyl formamide (DMF) tetrapeptide: L-Ala-L-Ala-L-Ala-L-Ala four C=O signals: 168.9, 171.6, 171.8, 173.8

More on these structures

Aldehydes and ketones

The first aldehyde is vanillin, which comes from the vanilla pod and gives the characteristic vanilla flavour in, for example, ice cream. Vanilla is the seed pod of a South American orchid. 'Vanilla essence' is made with synthetic vanillin and tastes slightly different because the vanilla pod contains other flavour components in small quantities. The second aldehyde is retinal. As you look at this structure your eyes use the light reaching them to interconvert *cis* and *trans* retinal in your retina to create nervous impulses (see also Chapter 27).

The two ketones are all flavour compounds too. The first, (-)-carvone, is the chief component (70%) of spearmint oil. Carvone is an interesting compound: in Chapter 14 you met the mirror-image isomers known as enantiomers, and (-)-carvone's mirror image, (+)-carvone, is the chief component (35%) of dill oil. Our taste can tell the difference, although an NMR machine can't and both carvones have *identical NMR spectra*. See Chapter 14 for more detail! The second ketone is 'raspberry ketone', which is largely responsible for the flavour of raspberries. It is entirely responsible for the flavour of some 'raspberry' foods. The signal for the aromatic carbon joined to OH is at 154.3 ppm (in the 100–150 ppm region because it is an unsaturated carbon atom joined to oxygen) and cannot possibly be confused with the ketone signal at 208.8 ppm. Both ketones have C=O shifts at about 200 ppm, and both lack any signals in the proton NMR of $\delta > 8$.

Acid derivatives

Lipoic acid uses its S–S bond in redox reactions (Chapter 42), while shikimic acid is an intermediate in the formation of compounds with benzene rings, such as phenylalanine, in living things (Chapter 42). Salicylic acid's acetate ester is aspirin, which is, of course, like the last example ibuprofen, a painkiller.

The first acid chloride is a popular reagent for the synthesis of acetate esters and you have seen its reactions in Chapter 10. We have chosen three cyclic anhydrides as examples because they are all related to an important reaction (the Diels–Alder reaction), which you will meet in Chapter 34.

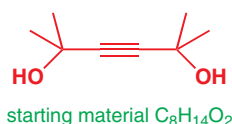
The first ester, methyl methacrylate, is a bulk chemical. It is the monomer whose polymerization gives Perspex, the rigid transparent plastic used in windows and roofs. The second ester is an important local anaesthetic used for minor operations.

One amide is the now-familiar DMF, but the other is a tetrapeptide and so contains one carboxylic acid group at the end and three amide groups. Although the four amino acids in this peptide are identical (alanine, Ala for short), the carbon NMR faithfully picks up four different C=O signals, all made different by being different distances from the end of the chain.

The distinction can be vital in structural problems. The symmetrical alkyne diol below cyclizes in acid with Hg(II) catalysis to a compound having, by proton NMR, the structural fragments shown. The product is unsymmetrical in that the two CMe₂ groups are still present, but they are now different. In addition, the chemical shift of the CH₂ group shows that it is next to C=O but not next to oxygen. This leaves us with two possible structures. One is an ester and one a ketone. The C=O shift is 218.8 ppm and so there is no doubt that the second structure is correct.

■ You need not, at this stage, worry about *how* the reaction works. It is more important that you realize how spectroscopy enables us to work out *what* has happened even before we have any idea *how*. Nonetheless, it is true that the second structure here also makes more sense chemically as the carbon skeleton is the same as in the starting material.

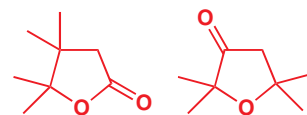
a reaction with an unknown product



the product is an isomer of C₈H₁₄O₂
¹H NMR shows these fragments:

[CMe₂, CMe₂, C=O, O, CH₂]

product might be one of these:



Distinguishing aldehydes from ketones is simple by proton NMR

Now look at the first two groups, the aldehydes and ketones. The two aldehydes have smaller carbonyl shifts than the two ketones, but they are too similar for this distinction to be reliable. What distinguishes the aldehydes very clearly is the characteristic proton signal for CHO at 9–10 ppm. So you should identify aldehydes and ketones by C=O shifts in carbon NMR and then separate the two by proton NMR.

● Aldehyde protons are characteristic

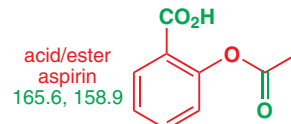
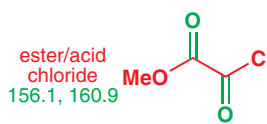
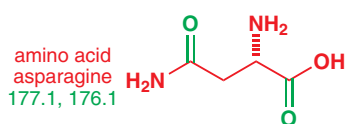
A proton at 9–10 ppm indicates an aldehyde.

Distinguishing acid derivatives by carbon NMR is difficult

Now examine the other panels on p. 409. The four carboxylic acids are all important biologically or medicinally. Their C=O shifts are very different *from each other* as well as from those of the aldehydes or ketones.

The next five compounds (two acid chlorides and three anhydrides) are all reactive acid derivatives, and the five esters and amides below them are all unreactive acid derivatives and yet the C=O shifts of all ten compounds fall in the same range. The C=O chemical shift is obviously *not* a good way to check on chemical reactivity.

What the carbon NMR fails to do is distinguish these types of acid derivative. There is more variation between the carboxylic acids on display than between the different classes of acid derivatives. This should be obvious if we show you some compounds containing two acid derivatives. Would you care to assign these signals?



➡ The relative reactivity of carboxylic acid derivatives was discussed in Chapter 10.

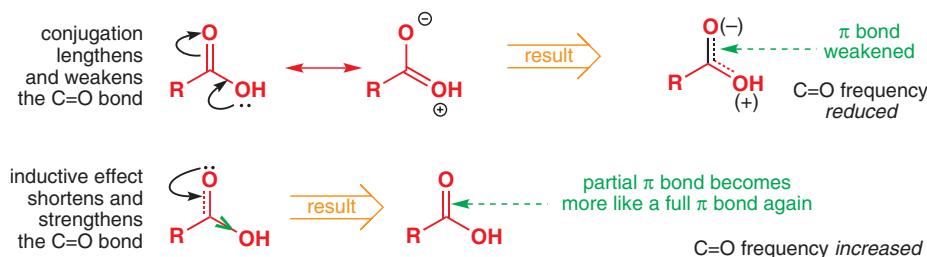
No, neither would we. In each case the difference between the carbonyl signals is only a few ppm. Although acid chlorides are extremely reactive in comparison with esters or amides, the electron deficiency at the carbon nucleus as measured by deshielding in the NMR spectrum evidently does not reflect this. Carbon NMR reliably distinguishes acid derivatives as a group from aldehydes and ketones as another group but it fails to distinguish even very reactive (for

example, acid chlorides) from very unreactive (for example, amides) acid derivatives. So how do we distinguish acid derivatives?

Acid derivatives are best distinguished by infrared

A much better measure is the difference in IR stretching frequency of the C=O group. We discussed this in Chapter 10 (p. 206), where we noted a competition between conjugation by lone-pair electron donation *into* the carbonyl from OCOR, OR, or NH₂ and inductive withdrawal *from* the C=O group because of the electronegativity of the substituent. Conjugation donates electrons into the π^* orbital of the π bond and so lengthens and weakens it. The C=O bond becomes more like a single bond and its stretching frequency moves towards the single-bond region, that is, it goes *down*. The inductive effect removes electrons from the π orbital and so shortens and strengthens the π bond. It becomes more like a full double bond and moves *up* in frequency.

➔ For a reminder of the distinction between conjugation and inductive effects, see Chapter 8, p. 176.



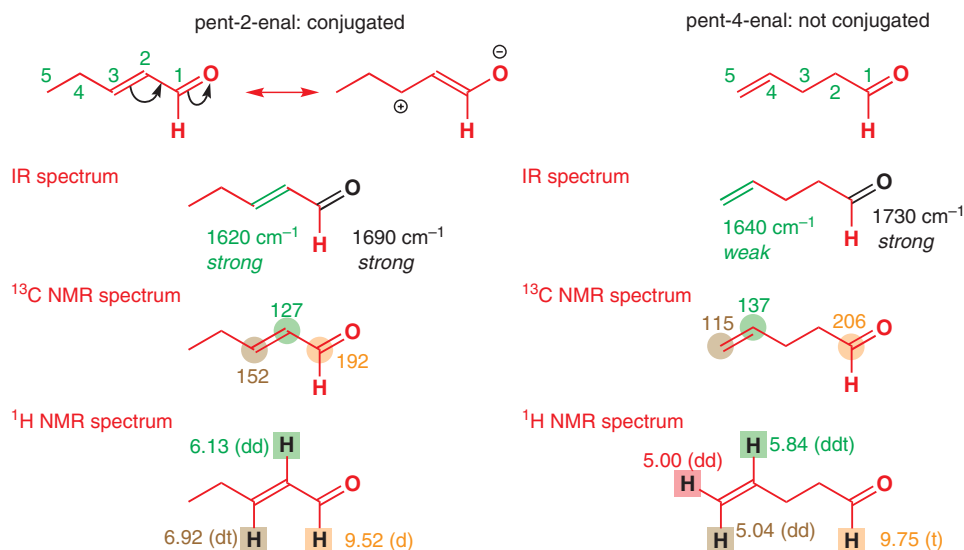
These effects are balanced in different ways according to the substituent. Chlorine is poor at lone-pair electron donation (its lone pair is in a large 3p orbital and overlaps badly with the 2p orbital on carbon) but strongly electron-withdrawing so acid chlorides absorb at high frequency, almost in the triple-bond region. Anhydrides have an oxygen atom between two carbonyl groups. Inductive withdrawal is still strong but conjugation is weak because the lone pairs are pulled both ways. Esters have a well-balanced combination with the inductive effect slightly stronger (oxygen donates from a compatible 2p orbital but is very electronegative and so withdraws electrons strongly as well). Finally, amides are dominated by conjugation as nitrogen is a much stronger electron donor than oxygen because it is less electronegative.

Acid chlorides	Anhydrides	Esters	Amides
inductive effect dominates 1815 cm ⁻¹	tug-of-war for lone pair: inductive effect dominates two peaks: ~1790, 1810 cm ⁻¹	inductive effect slightly dominates 1745 cm ⁻¹	conjugation strongly dominates ~ 1650 cm ⁻¹

■ The two peaks for anhydrides are the symmetrical and anti-symmetrical stretches for the two C=O groups; see Chapter 3, p. 70.

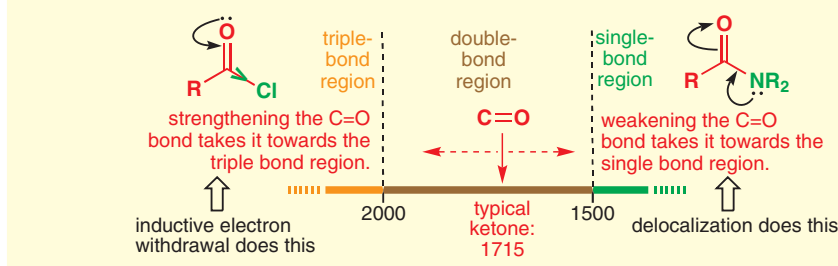
Conjugation with π electrons or lone pairs affects IR C=O stretches

We need to see how conjugation works when it is with a π bond rather than with a lone pair. This will make the concept more general as it will apply to aldehydes and ketones as well as to acid groups. How can we detect whether an unsaturated carbonyl compound is conjugated or not? Well, compare these two unsaturated aldehydes.



The key differences are the frequency of the C=O stretch (lowered by 40 cm^{-1} by conjugation) and the strength (that is, the intensity) of the C=C stretch (increased by conjugation) in the IR. In the ^{13}C NMR, C3 in the conjugated enal is moved out of the alkene region just into the carbonyl region, showing how electron-deficient this carbon atom must be. In the proton NMR there are many effects but the downfield shift of the protons on the alkene, especially C3 (again!), is probably the most helpful.

• Summary of the effects of substituents on C=O stretching frequency



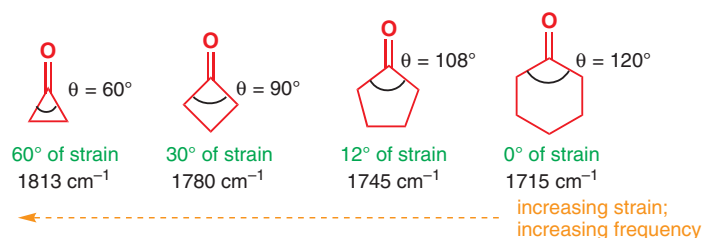
Because the infrared carbonyl frequencies follow such a predictable pattern, it is possible to make a simple list of correlations using just three factors. Two are the ones we have been discussing—conjugation (frequency-lowering) and the inductive effect (frequency-raising). The third is the effect of small rings and this we next need to consider in a broader context.

Small rings introduce strain inside the ring and higher s character outside it

■ The three-membered ring is, of course, flat. The others are not. Even the four-membered ring is slightly puckered, the five and especially the six-membered rings more so. This is all discussed, along with analysis of ring strain, in Chapter 16.

Cyclic ketones can achieve the perfect 120° angle at the carbonyl group only if the ring is at least six-membered. The smaller rings are 'strained' because the orbitals have to overlap at a less than ideal angle.

For a four-membered ring, the actual angle is 90° , so there is $120^\circ - 90^\circ = 30^\circ$ of strain at the carbonyl group. The effects of this strain on five-, four-, and three-membered rings are shown here.



But why should strain raise the frequency of a carbonyl group? It is evidently shortening and strengthening the C=O bond as it moves it towards the triple-bond region (higher frequency), not towards the single-bond region (lower frequency). In a six-membered ring, the sp² orbitals forming the σ framework around the carbonyl group can overlap perfectly with the sp³ orbitals on neighbouring carbon atoms because the orbital angle and the bond angle are the same. In a four-membered ring the orbitals do not point towards those on the neighbouring carbon atoms, but point too far out, effectively forcing the bonds to be bent and lowering the degree of overlap.

Ideally, we should like the orbitals to have an angle of 90° as this would make the orbital angle the same as the bond angle. In theory it *would* be possible to have a bond angle of 90° if we used pure p orbitals instead of sp² hybrid orbitals. The diagram in the margin shows this hypothetical situation. If we did this, we should leave a pure s orbital for the σ bond to oxygen. This extreme is not possible, but a compromise is. *Some* more p character goes into the ring bonds—maybe they become s^{0.8}p^{3.2}—so that they can approach the 90° angle needed, and the same amount of extra s character goes into the σ bond to oxygen. The more s character there is in the orbital, the shorter it gets as s orbitals are much smaller than p orbitals.

Simple calculations of C=O stretching frequencies in IR spectra

The best way is to relate all our carbonyl frequencies to those for saturated ketones (1715 cm⁻¹). We can summarize what we have just learned in a table.

Notice in this simple table (for full details you should refer as usual to a specialist book) that the adjustment '30 cm⁻¹' appears quite a lot (−30 cm⁻¹ for both alkene and aryl, for example), that the increment for small rings is 35 cm⁻¹ each time (30 to 65 cm⁻¹ and then 65 to 100 cm⁻¹), and that the extreme effects of Cl and NH₂ are +85 and −85 cm⁻¹, respectively. These effects are additive. If you want to estimate the C=O frequency of a proposed structure, just add or subtract all the adjustments to 1715 cm⁻¹ and you will get a reasonable result.

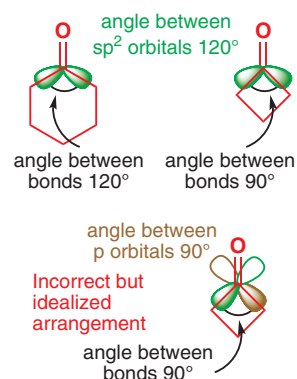
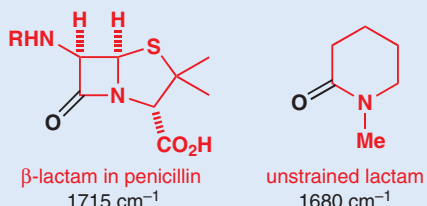
Effects of substituents on IR carbonyl frequencies

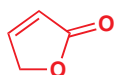
Effect	Group	C=O stretch, cm ⁻¹	Frequency change ^a , cm ⁻¹
inductive effect	Cl	1800	+ 85
	OCOR	1765, 1815	+ 50, +100
	OR	1745	+ 30
	H	1730	+ 15
conjugation	C=C	1685	−30
	aryl	1685	−30
	NH ₂	1630	−85
ring strain	five-membered ring	1745	+ 30
	four-membered ring	1780	+ 65
	three-membered ring	1815	+ 100

^aDifference between stretching frequency of C=O and stretching frequency of a typical saturated ketone (1715 cm⁻¹).

Lactam C=O stretching frequencies

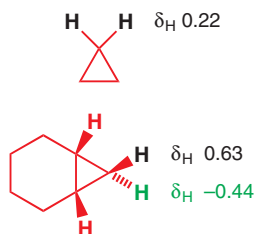
A further good example is the difference between C=O stretching frequencies in cyclic amides, or lactams. The penicillin class of antibiotics all contain a four-membered ring amide known as a β-lactam. The carbonyl stretching frequency in these compounds is way above the 1680 cm⁻¹ of the six-membered lactam, which is what you might expect for an unstrained amide.





Try this out with the five-membered unsaturated (and conjugated) lactone (cyclic ester) in the margin. We must add 30 cm^{-1} for the ester, subtract 30 cm^{-1} for the double bond, and add 30 cm^{-1} for the five-membered ring. Two of those cancel out, leaving just $1715 + 30 = 1745\text{ cm}^{-1}$. These compounds absorb at $1740\text{--}1760\text{ cm}^{-1}$. Not bad!

NMR spectra of alkynes and small rings

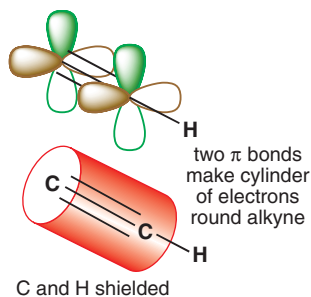
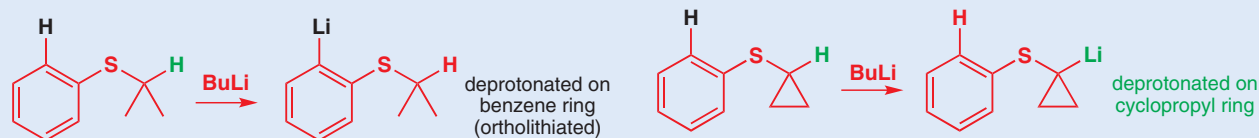


This idea that small rings have more p character in the ring and more s character outside the ring also explains the effects of small rings on proton NMR shifts. These hydrogens, particularly on three-membered rings, resonate at unusually high fields, between 0 and 1 ppm in cyclopropanes instead of the 1.3 ppm expected for CH_2 groups, and may even appear at negative δ values. High p character in the framework of small rings also means high s character in C–H bonds outside the ring and this will mean shorter bonds, greater shielding, and small δ values.

Three-membered rings and alkynes

You have also seen the same argument used in Chapter 8 to justify the unusual acidity of C–H protons on triple bonds (such as alkynes and HCN), and alluded to in Chapter 3 to explain the stretching frequency of the same C–H bonds. Like alkynes, three-membered rings are also unusually easy to deprotonate in base.

Here is an example where deprotonation occurs at a different site in two compounds identical except for a C–C bond closing a three-membered ring. The first is an ortholithiation of the type discussed in Chapter 24.

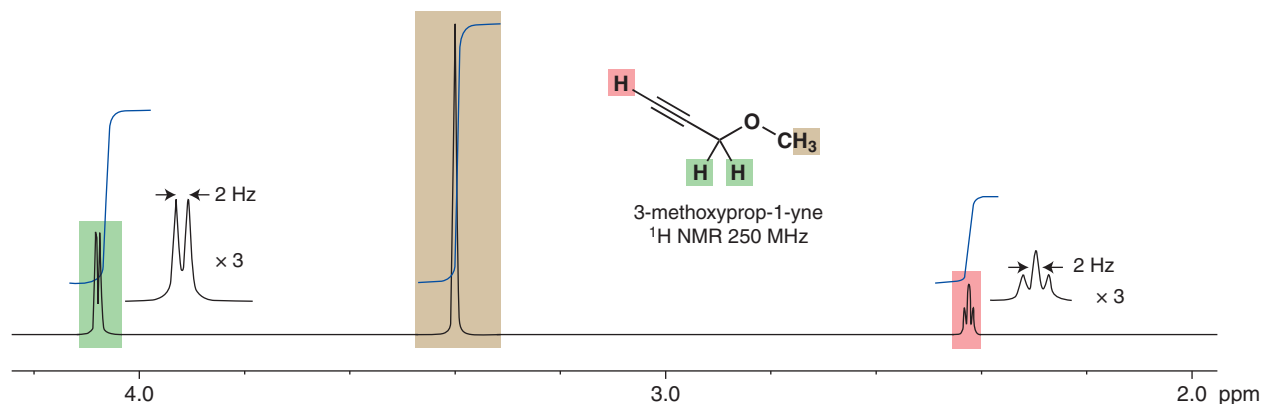


Now what about the NMR spectra of alkynes? By the same argument, protons on alkynes ought to appear in the NMR at quite high field because the C atom is sp hybridized, so it makes its σ bonds with sp orbitals (i.e. 50% s character). Protons on a typical alkene have δ_H about 5.5 ppm, while the proton on an alkyne comes right in the middle of the protons on saturated carbons at about δ_H 2–2.5 ppm. This is rather a large effect just for increased s character and some of it is probably due to better shielding by the triple bond, which surrounds the linear alkyne with π bonds without a nodal plane.

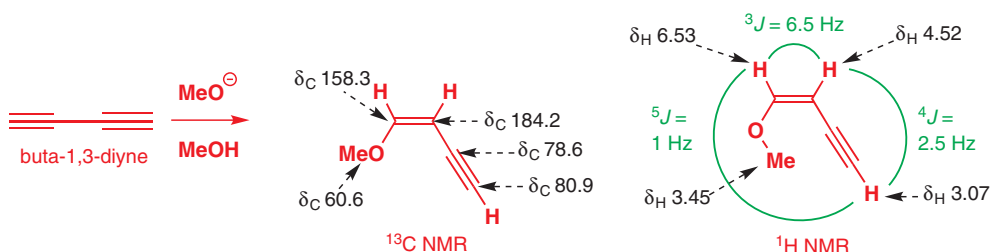
This means that the carbon atoms also appear at higher field than expected, not in the alkene region but from about δ_C 60–80 ppm. The s character argument is important, however, because shielding can't affect IR stretching frequencies, yet $\text{C}\equiv\text{C-H}$ stretches are strong and at about 3300 cm^{-1} , just right for a strong C–H bond.

A simple example is the ether 3-methoxyprop-1-yne. Integration alone allows us to assign the spectrum and the ^1H signal at 2.42 ppm, the highest field signal, is clearly the alkyne proton. Notice also that it is a triplet and that the OCH_2 group is a doublet. This $^4J_{\text{HH}}$ is small (about 2 Hz) and, although there is nothing like a letter 'W' in the arrangement of the bonds, coupling of this kind is often found in alkynes.

➔ In Chapter 13, p. 296, you saw that bonds aligned in a 'W' arrangement can give rise to a small $^4J_{\text{HH}}$ coupling.



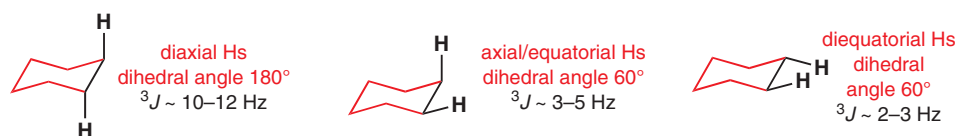
A more interesting example comes from the base-catalysed addition of methanol to buta-1,3-diyne (diacetylene). The compound formed has one double and one triple bond and the ^{13}C NMR shows clearly the greater deshielding of the double bond.



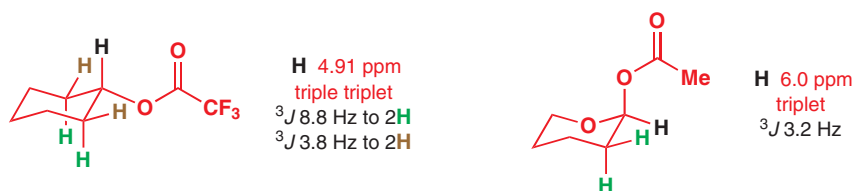
You may have noticed that we have drawn the double bond with the *cis* (Z) configuration. We know that this is true because of the proton NMR, which shows a 6.5 Hz coupling between the two alkene protons (much too small for a *trans* coupling; see p. 295). There is also the longer-range coupling ($^4J = 2.5 \text{ Hz}$) just described and even a small very long-range coupling ($^5J = 1 \text{ Hz}$) between the alkyne proton and the terminal alkene proton.

Proton NMR distinguishes axial and equatorial protons in cyclohexanes

Coupling is a through-bond phenomenon, as we know from the couplings in *cis* and *trans* alkenes, where *trans* alkenes have much larger coupling constants as their orbitals are perfectly parallel. Another case of perfectly parallel orbitals occurs with *trans*-diaxial protons in cyclohexanes. Typical coupling constants are 10–12 Hz for *trans*-diaxial protons, but much smaller (2–5 Hz) for axial/equatorial and equatorial/equatorial protons.



This makes assignment of conformation easy. The simple ester below has a triple triplet for the black H, with two large coupling constants (8.8 Hz) that must be to axial protons (green) and two small coupling constants (3.8 Hz) that must be to equatorial Hs (brown). This is possible only if the black H is axial and the ester group must therefore be equatorial. The acetal ester on the right is very different: it is a simple triplet with two small coupling constants (3.2 Hz), which is too small for an axial/axial coupling. The only possibility therefore is that the black proton is equatorial, and one of the 3.2 Hz couplings is to its equatorial neighbour, and the other to its axial neighbour. The ester group must be axial in this compound.



Interactions between different nuclei can give enormous coupling constants

We have looked at coupling between hydrogen atoms and you may have wondered why we have ignored coupling between other NMR active nuclei. Why does ^{13}C not cause similar couplings? In this section we are going to consider not only couplings between the same kind of nuclei,

➔ Coupling in alkenes is explained on p. 295.

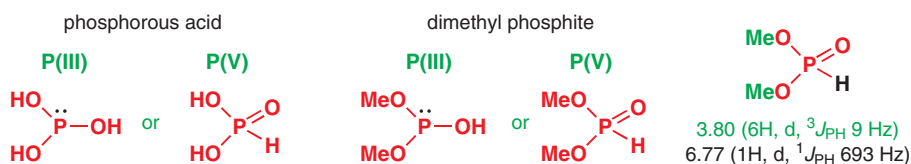
➔ Proton–proton coupling in alkenes is discussed in Chapter 13 and the conformation of cyclohexanes is discussed in Chapter 16. The Karplus relationship, explaining precisely what affects the couplings in cyclohexanes, is discussed in Chapter 31.

➔ You will see in Chapter 31 why the ester group might prefer to be axial in this compound.

such as two protons, called **homonuclear coupling**, but also coupling between different nuclei, such as a proton and a fluorine atom or ^{13}C and ^{31}P , called **heteronuclear coupling**.

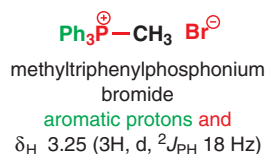
Two nuclei are particularly important, ^{19}F and ^{31}P , since many organic compounds contain these elements and both are at essentially 100% natural abundance and have spin $I = 1/2$. We shall start with organic compounds that have just one of these nuclei and see what happens to both the ^1H and the ^{13}C spectra. In fact, it is easy to find a ^{19}F or a ^{31}P atom in a molecule because these elements couple to all nearby carbon and hydrogen atoms. Since they can be directly bonded to either, 1J coupling constants such as $^1J_{\text{CF}}$ or $^1J_{\text{PH}}$ become possible, as well as the more ‘normal’ couplings such as $^2J_{\text{CF}}$ or $^3J_{\text{PH}}$, and these 1J coupling constants can be enormous.

We shall start with a simple phosphorus compound, the dimethyl ester of phosphorous acid (H_3PO_3). There is an uncertainty about the structure of both the acid and its esters. They could exist as P(III) compounds with a lone pair of electrons on phosphorus, or as P(V) compounds with a $\text{P}=\text{O}$ double bond.



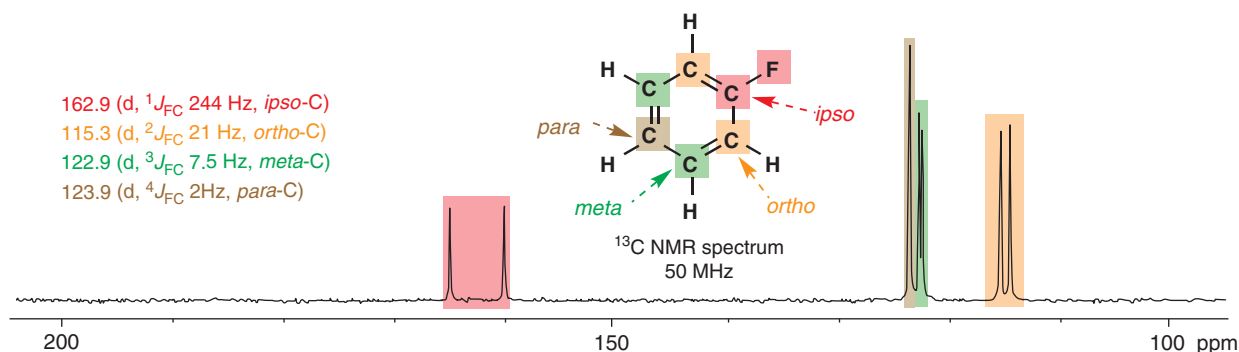
In fact, dimethyl phosphite has a ^1H doublet with the amazing coupling constant of 693 Hz: on a 250 MHz machine the two lines are over 2 ppm apart and it is easy to miss that they are two halves of the same doublet. This can only be a $^1J_{\text{PH}}$ as it is so enormous. The compound has to have a $\text{P}-\text{H}$ bond and the P(V) structure is correct. The coupling to the protons of the methyl group is much smaller but still large for a three-bond coupling ($^3J_{\text{PC}}$ of 18 Hz).

Next, consider the phosphonium salt you met at the end of Chapter 11 for use in the Wittig reaction, turning aldehydes and ketones to alkenes. It has a $^2J_{\text{PH}}$ of 18 Hz. There is no doubt about this structure—it is just an illustration of coupling to phosphorus. There is coupling to phosphorus in the carbon spectrum too: the methyl group appears at δ_{C} 10.6 ppm with a $^1J_{\text{PC}}$ of 57 Hz, somewhat smaller than typical $^1J_{\text{PH}}$. We haven't yet talked about couplings to ^{13}C : we shall now do so.



Coupling in carbon NMR spectra

We shall use coupling with fluorine to introduce this section. Fluorobenzenes are good examples because they have a number of different carbon atoms all coupled to the fluorine atom.

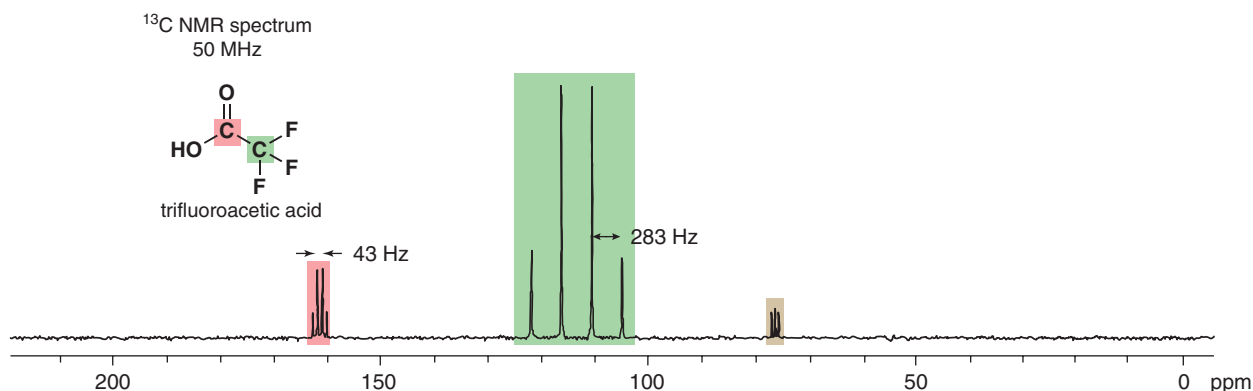


■ *Ipso* can join the list (*ortho*, *meta*, *para*) of trivial names for positions on a substituted benzene ring. The *ipso* carbon is the one directly attached to a substituent.

The carbon directly joined to fluorine (the *ipso* carbon) has a very large $^1J_{\text{CF}}$ value of about 250 Hz. More distant coupling is evident too: all the carbons in the ring couple to the fluorine in PhF with steadily diminishing J values as the carbons become more distant.

Trifluoroacetic acid is an important strong organic acid (Chapter 8) and a good solvent for ^1H NMR. The carbon atom of the CF_3 group is coupled equally to all the three fluorines and so appears as a quartet with a large $^1J_{\text{CF}}$ of 283 Hz, about the same as in PhF. Even the carbonyl

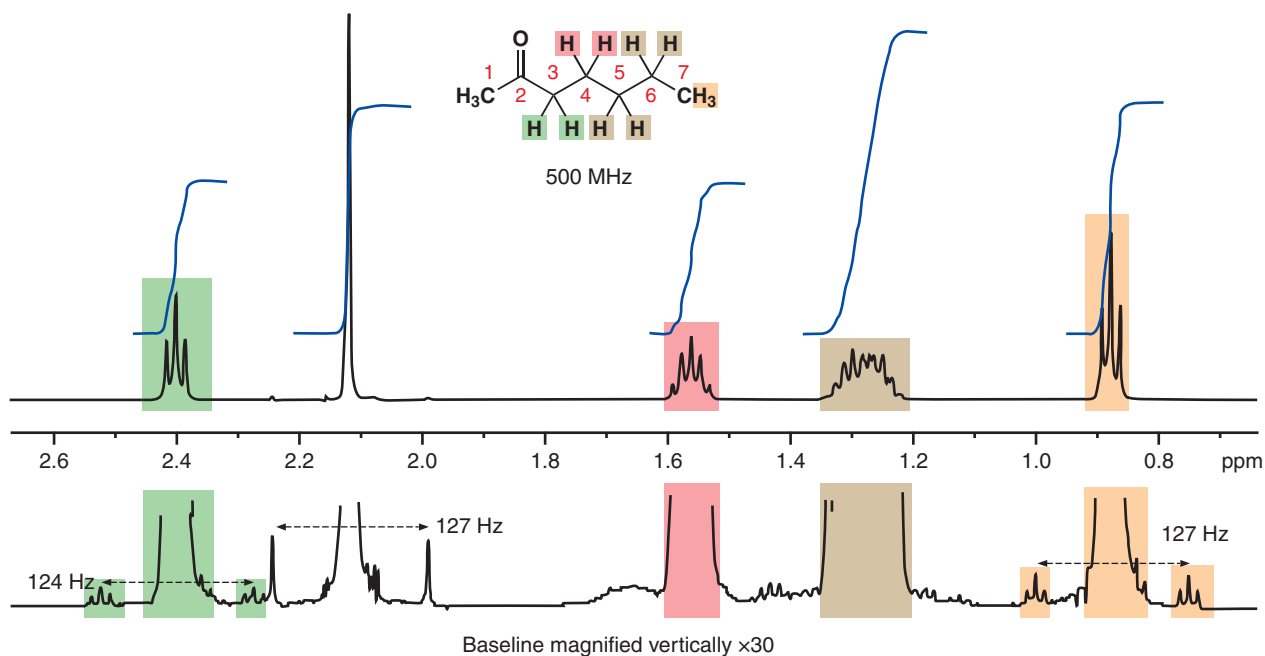
group is also a quartet, although the coupling constant is much smaller ($^2J_{\text{CF}}$ is 43 Hz). Notice too how far downfield the CF_3 carbon atom is!



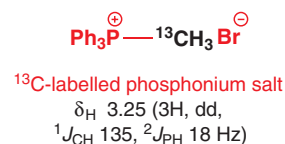
Coupling between protons and ^{13}C

In view of all this, you may ask why we don't apparently see couplings between ^{13}C and ^1H in either carbon or proton spectra. In proton spectra the answer is simple: we don't see coupling to ^{13}C because of the low abundance (1.1%) of ^{13}C . Most protons are bonded to ^{12}C : only 1.1% of protons are bonded to ^{13}C . If you look closely at proton spectra with very flat baselines, you may see small peaks either side of strong peaks at about 0.5% peak height. These are the ^{13}C 'satellites' for those protons that are bonded to ^{13}C atoms.

As an example, look again at the 500 MHz ^1H NMR spectrum of heptan-2-one that we saw on p. 294. When the baseline of this spectrum is vertically expanded, the ^{13}C satellites may be seen. The singlet due to the methyl protons is actually in the centre of a tiny doublet due to the 1% of protons coupling to ^{13}C . Similarly, each of the triplets in the spectrum is flanked by two tiny triplets. The two tiny triplets on either side make up a doublet of triplets with a large 1J coupling constant to the ^{13}C (around 130 Hz) and smaller 3J coupling to the two equivalent protons.



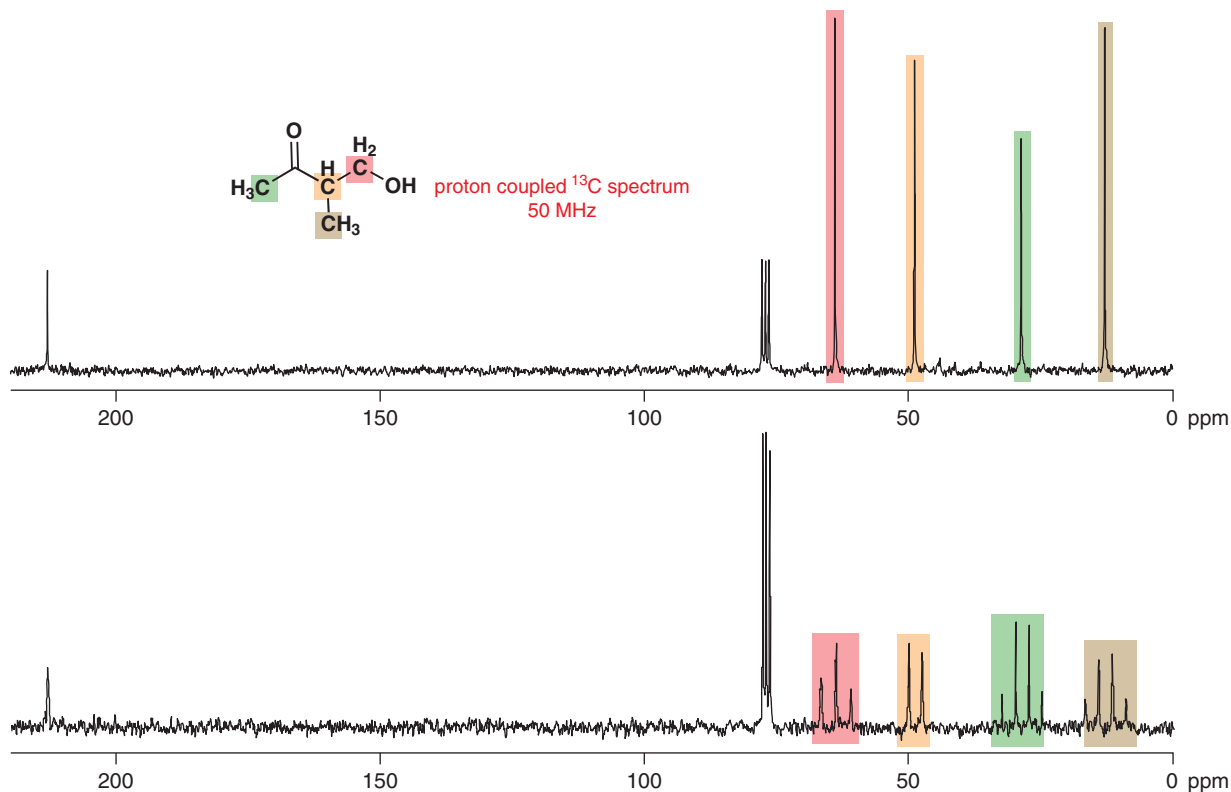
^{13}C satellites are usually lost in the background noise of the spectrum and need concern us no further. You do, however, see coupling in the ^1H NMR spectrum with compounds deliberately labelled with ^{13}C because the ^{13}C abundance can then approach 100%. The same Wittig reagent we saw a moment ago shows a 3H doublet of doublets with the typically enormous $^1J_{\text{CH}}$ of 135 Hz when labelled with pure ^{13}C in the methyl group.



But this begs the question—where is the 135 Hz coupling in the ^{13}C NMR? Surely we should see this coupling to the protons in the ^{13}C NMR spectrum too?

Why is there no coupling to protons in normal ^{13}C NMR spectra?

We get the singlets consistently seen in carbon spectra because of the way we record the spectra. The values of $^1J_{\text{CH}}$ are so large that, if we recorded ^{13}C spectra with all the coupling constants, we would get a mass of overlapping peaks. When run on the same spectrometer, the frequency at which ^{13}C nuclei resonate turns out to be about a quarter of that of the protons. Thus a '400 MHz machine' (remember that the magnet strength is usually described by the frequency at which the protons resonate) gives ^{13}C spectra at 100 MHz. Coupling constants ($^1J_{\text{CH}}$) of 100–250 Hz would cover 2–5 ppm and a CH_3 group with $^1J_{\text{CH}}$ of about 125 Hz would give a quartet covering nearly 8 ppm (see the example on the previous page).



Since the proton-coupled ^{13}C spectrum can so easily help us to distinguish CH_3 , CH_2 , CH , and quaternary carbons, you might wonder why they are not used more. The above example was chosen very carefully to illustrate proton-coupled spectra at their best. Unfortunately, this is not a typical example. More usually, the confusion from overlapping peaks makes this just not worthwhile. So ^{13}C NMR spectra are recorded while the whole 10 ppm proton spectrum is being irradiated with a secondary radio frequency source. The proton energy levels are equalized by this process and all coupling disappears. Hence the singlets we are used to seeing.

For the rest of this chapter we shall not be introducing new theory or new concepts; we shall be applying what we have told you to a series of examples where spectroscopy enables chemists to identify compounds.

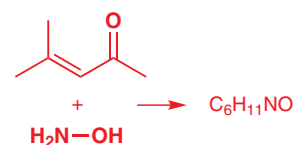
Identifying products spectroscopically

An ambiguous reaction product

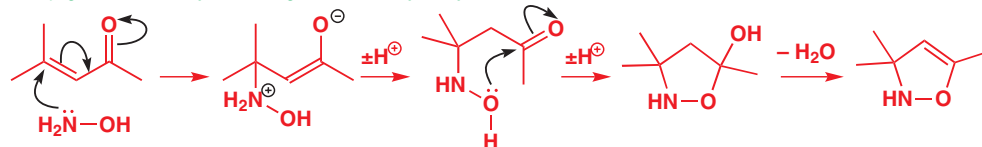
➡ This was the case of diazonamide A (p. 45).

In Chapter 3 we gave an example of a compound which was misidentified because an O atom and an N atom were mistaken for one another, even in the X-ray crystal structure.

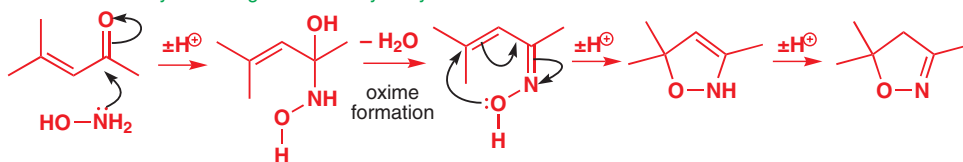
Another famous case of ambiguity between structures containing O or N arises in the identification of the product of addition of hydroxylamine (NH_2OH) to a simple enone. This condensation reaction gives a compound with the formula $\text{C}_6\text{H}_{11}\text{NO}$. But what is its structure? We can first of all think about what we expect to happen: it is not always necessary to do this in order to identify a structure, but it can help. Nitrogen is more nucleophilic than oxygen so we might expect it to add first. But will it add directly to the carbonyl group or in the conjugate fashion we shall describe in Chapter 22? Either way, an intermediate will be formed that can cyclize.



conjugate addition by the nitrogen atom of hydroxylamine



direct addition by the nitrogen atom of hydroxylamine



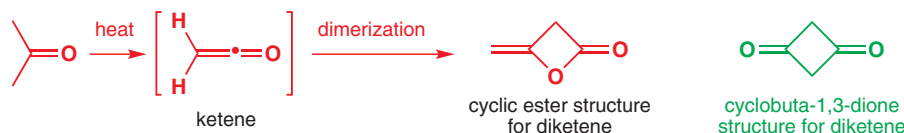
The two possible isomeric products were once the subject of a long-running controversy, but with IR and proton NMR spectra of the product, doubt vanished. The IR showed no NH stretch. The NMR showed no alkene proton but did have a CH_2 group at 2.63 ppm. Only the second structure is possible.

We need to look now at a selection of problems of different kinds to show how the various spectroscopic methods can cooperate in structure determination.

Reactive intermediates can be detected by spectroscopy

Some intermediates proposed in reaction mechanisms look so unlikely that it is comforting if they can be isolated and their structure determined. We feel more confident in proposing an intermediate if we are sure that it can really be made. Of course, this is not necessarily evidence that the intermediate is actually formed during reactions and it certainly does *not* follow that the failure to isolate a given intermediate disproves its involvement in a reaction. We shall use ketene as an example.

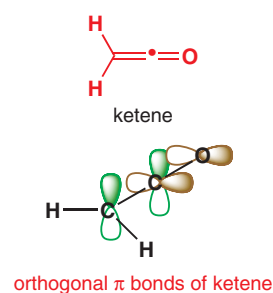
Ketene looks pretty unlikely! It is $\text{CH}_2=\text{C}=\text{O}$ with two π bonds ($\text{C}=\text{C}$ and $\text{C}=\text{O}$) to the same carbon atom. The orbitals for these π bonds must be orthogonal because the central carbon atom is sp hybridized with two linear σ bonds and two p orbitals at right angles both to the σ bonds and to each other. Can such a molecule exist? When acetone vapour is heated to very high temperatures (700–750 °C) methane is given off and ketene is supposed to be the other product. What is isolated is a ketene dimer ($\text{C}_4\text{H}_4\text{O}_2$) and even the structure of this is in doubt as two reasonable structures can be written.



The spectra fit the ester structure well, but not the more symmetrical diketone structure at all. There are *three* types of proton (cyclobuta-1,3-dione would have just *one*), with allylic coupling between one of the protons on the double bond and the CH_2 group in the ring. The carbonyl group has the shift (185 ppm) of an acid derivative (not that of a ketone, which would be about 200 ppm) and all four carbons are different.

Do not be concerned about the details of the mechanisms: note that we have used the ' $\pm \text{H}^+$ ' shorthand introduced in Chapter 11, and have abbreviated the mechanism where water is eliminated and the oxime formed—the full mechanism of imine (and oxime) formation can be found in Chapter 11, p. 229. In this chapter, we are much more concerned just with the structure of the products.

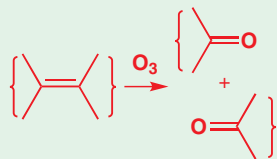
We used this logic in Chapter 15: carbocations were proposed as intermediates in $\text{S}_{\text{N}}1$ reactions long before they were observed spectroscopically, but it was reassuring to be able to see them by NMR once appropriate conditions were devised (see p. 335).



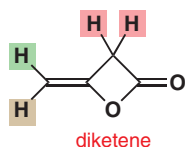
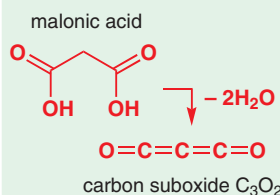
The structure of ketene is analogous to that of *allene*, discussed in Chapter 7, p. 146. Ketene is isoelectronic (p. 354) with CO_2 and azide, N_3^- .

■ Ozonolysis or ozonation

is the cleavage of an alkene by ozone (O_3). The reaction and its mechanism are discussed in Chapters 19 and 34: the only point to note now is that ozone is a powerful oxidant and cleaves the alkene to make two carbonyl compounds. Again, in this chapter we are concerned only with the structure of the products and how these can be determined.



■ Malonic anhydride cannot be made directly from malonic acid because attempted dehydration of the acid leads to the exotic molecule carbon suboxide C_3O_2 .



1H NMR spectrum:

4.85 (1H, narrow t, $J \sim 1$)

4.51 (1H, s)

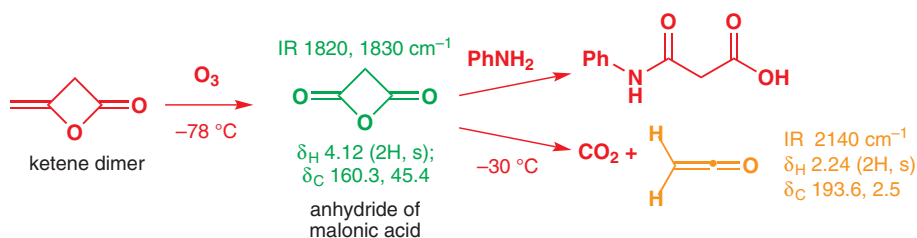
3.90 (2H, d, $J \sim 1$)

^{13}C NMR spectrum:

185.1, 147.7, 67.0, 42.4

Ozonolysis of ketene dimer gives a very unstable compound that can be observed only at low temperatures ($-78^\circ C$ or below). It has two carbonyl bands in the IR and reacts with amines to give amides, so it looks like an anhydride (Chapter 10). Can it be the previously unknown cyclic anhydride of malonic acid?

The two carbonyl bands are of high frequency, as would be expected for a four-membered ring—using the table on p. 413 we estimate $1715 + 50\text{ cm}^{-1}$ (for the anhydride) + 65 cm^{-1} (for the four-membered ring) = 1830 cm^{-1} . Both the proton and the carbon NMR are very simple: just a 2H singlet at 4.12 ppm, shifted downfield by two carbonyls, a $C=O$ group at 160 ppm, right for an acid derivative, and a saturated carbon shifted downfield but not as much as a CH_2O group.



All this is reasonably convincing, and is confirmed by allowing the anhydride to warm to $-30^\circ C$, at which temperature it loses CO_2 (detected by the ^{13}C peak at 124.5 ppm) and gives another unstable compound with the strange IR frequency of 2140 cm^{-1} . Could this be monomeric ketene? It's certainly not either of the possible ketene dimers as we know what their spectra are like, and this is quite different: just a 2H singlet at 2.24 ppm and ^{13}C peaks at 194.0 and, remarkably, 2.5 ppm. It is indeed monomeric ketene.

Squares and cubes: molecules with unusual structures

Some structures are interesting because we believe they can tell us something fundamental about the nature of bonding while others are a challenge because many people argue that they cannot be made. What do you think are the prospects of making cyclobutadiene, a conjugated four-membered ring, or the hydrocarbons tetrahedrane and cubane, which have, respectively, the shapes of the perfectly symmetrical Euclidean solids, the tetrahedron and the cube?

With four electrons, cyclobutadiene is anti-aromatic—it has $4n$ instead of $4n + 2$ electrons. You saw in Chapter 7 that cyclic conjugated systems with $4n$ electrons (cyclooctatetraene, for example) avoid being conjugated by puckering into a tub shape. Cyclobutadiene cannot do this: it must be more or less planar, and so we expect it to be very unstable. Tetrahedrane has four fused three-membered rings. Although the molecule is tetrahedral in shape, each carbon atom is nowhere near a tetrahedron, with three bond angles of 60° . Cubane has six fused four-membered rings and is again highly strained.

In fact, cubane has been made, cyclobutadiene has a fleeting existence but can be isolated as an iron complex, and a few substituted versions of tetrahedrane have been made. The most convincing evidence that you have made any of these three compounds would be the extreme simplicity of the spectra. Each has only one kind of hydrogen and only one kind of carbon. They all belong to the family $(CH)_n$.

Cubane has a molecular ion in the mass spectrum at 104, correct for C_8H_8 , only CH stretches in the IR at 3000 cm^{-1} , a singlet in the proton NMR at 4.0 ppm, and a single line in the carbon



cyclobutadiene



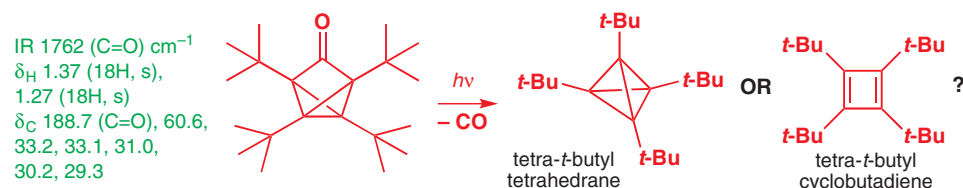
tetrahedrane



cubane

NMR at 47.3 ppm. It is a very symmetrical molecule and a stable one in spite of all those four-membered rings.

Stable compounds with a cyclobutadiene and a tetrahedrane core can be made if each hydrogen atom is replaced by a *t*-butyl group. The very large groups round the edge of the molecule repel each other and hold the inner core tightly together. Now another difficulty arises—it is rather hard to tell the compounds apart. They both have four identical carbon atoms in the core and four identical *t*-butyl groups round the edge. The starting material for a successful synthesis of both was the tricyclic ketone below identified by its strained C=O stretch and partly symmetrical NMR spectra. When this ketone was irradiated with UV light (indicated by '*hν*' in the scheme), carbon monoxide was evolved and a highly symmetrical compound (*t*-BuC)₄ was formed. But which compound was it?



➡ You can read more about the synthesis of cubane in Chapter 36, when we discuss the rearrangement reactions that were used to make it.

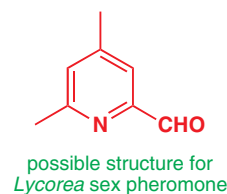
The story is made more complicated (but in the end easier!) by the discovery that this compound on heating turned into another very similar compound. There are only two possible structures for (*t*-BuC)₄, so clearly one compound must be the tetrahedrane and one the cyclobutadiene. The problem simplifies with this discovery because it is easier to distinguish two possibilities when you can make comparisons between two sets of spectra. Here both compounds gave a molecular ion in the mass spectrum, neither had any interesting absorptions in the IR, and the proton NMRs could belong to either compound as they simply showed four identical *t*-Bu groups. So did the carbon NMR, of course, but it showed the core too. The first product had only saturated carbon atoms, while the second had a signal at 152.7 ppm for the unsaturated carbons. The tetrahedrane is formed from the tricyclic ketone on irradiation but it isomerizes to the cyclobutadiene on heating.

Identifying compounds from nature

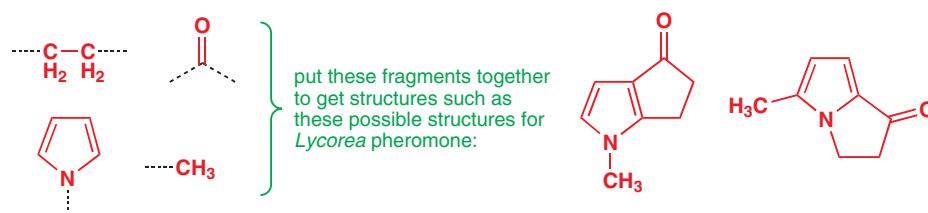
The next molecules we need to know how to identify are those discovered from nature—natural products. These often have biological activity and many useful medicines have been discovered this way. We shall look at a few examples from different fields. The first is the sex pheromone of the Trinidad butterfly *Lycorea ceres ceres*. The male butterflies start courtship by emitting a tiny quantity of a volatile compound. Identification of this type of compound is very difficult because of the minute amounts available but this compound was crystallized and gave enough for a mass spectrum and an IR. The highest peak in the mass spectrum was at 135. This is an odd number so we might have one nitrogen atom and a possible composition of C₈H₉ON. The IR showed a carbonyl peak at 1680 cm⁻¹. With only this meagre information, the first proposals were for a pyridine aldehyde.

Eventually a little more compound (6 mg!) was available and a proton NMR spectrum was run. This showed at once that this structure was wrong. There was no aldehyde proton and only one methyl group. More positive information was the pair of triplets showing a –CH₂CH₂– unit between two electron-withdrawing groups (N and C=O?) and the pair of doublets for neighbouring protons on an aromatic ring, although the chemical shift and the coupling constant are both rather small for a benzene ring.

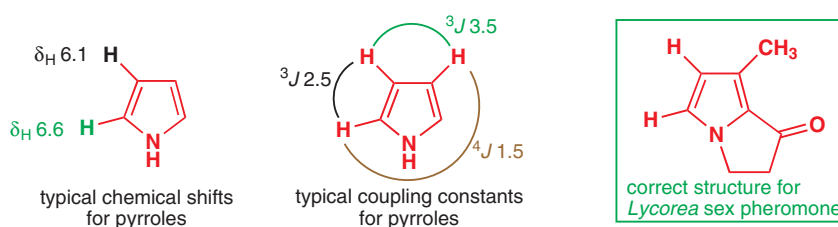
If we look at what we have got so far, we see that we have accounted for four carbon atoms in the methyl and carbonyl groups and the –CH₂CH₂– unit. This leaves only four carbon atoms for the aromatic ring. We must use nitrogen too as the only possibility is a pyrrole ring. Our fragments are now those shown below (the black dotted lines show joins to another fragment). These account for all the atoms in the molecule and suggest structures such as these.



➡ Pyrrole was introduced in Chapter 7, p. 162.



Now we need to use the known chemical shifts and coupling constants for these sorts of molecules. An N–Me group would normally have a larger chemical shift than 2.2 ppm so we prefer the methyl group on a carbon atom of the pyrrole ring. Typical shifts and coupling constants around pyrroles are shown below. Chemists do not, of course, remember these numbers; we look them up in tables. Our data, with chemical shifts of 6.09 and 6.69 ppm and a coupling constant of 2.5 Hz, clearly favour hydrogen atoms in the 2 and 3 positions, and suggest this structure for the sex pheromone, which was confirmed by synthesis and is now accepted as correct.



Tables

The final section of this chapter contains some tables of NMR data, which we hope you will find useful in solving problems. In Chapter 13 there were a few guides to chemical shift—summaries of patterns that you might reasonably be expected to remember. But we have left the main selections of hard numbers—tables that *you are not expected to remember*—until now. There are a few comments to explain the tables, but you will probably want to use this section as reference rather than bedtime reading. The first four tables give detailed values for various kinds of compounds and the final table gives a simple summary. We hope that you will find this last table particularly useful.

Effects of electronegativity

This table shows how the electronegativity of the atom attached directly to a methyl group affects the shifts of the CH₃ protons (δ_H) and the CH₃ carbon atom (δ_C) in their NMR spectra.

Chemical shifts of methyl groups attached to different atoms

Element	Electronegativity	Compound	δ_H , ppm	δ_C , ppm
Li	1.0	CH ₃ –Li	–1.94	–14.0
Si	1.9	CH ₃ –SiMe ₃	0.0	0.0
I	2.7	CH ₃ –I	2.15	–23.2
S	2.6	CH ₃ –SMe	2.13	18.1
N	3.1	CH ₃ –NH ₂	2.41	26.9
Cl	3.2	CH ₃ –Cl	3.06	24.9
O	3.4	CH ₃ –OH	3.50	50.3
F	4.0	CH ₃ –F	4.27	75.2

Effects of functional groups

Many substituents are more complicated than just a single atom and electronegativity is only part of the story. We need to look at all the common substituents and see what shifts they cause relative to the CH skeleton of the molecule. Our zero really ought to be at about 0.9 ppm for protons and at 8.4 ppm for carbon, that is, where ethane ($\text{CH}_3\text{--CH}_3$) resonates, and not at the arbitrary zero allocated to Me_4Si . In the table below we give such a list. The reason for this is that the shifts (from Me_4Si) themselves are not additive but the shift differences (from 0.9 or 8.4 ppm) are.

Chemical shifts of methyl groups bonded to functional groups

	Functional group	Compound	δ_{H} , ppm	$\delta_{\text{H}} - 0.9$, ppm	δ_{C} , ppm	$\delta_{\text{C}} - 8.4$, ppm
1	silane	Me₄Si	0.0	−0.9	0.0	−8.4
2	alkane	Me–Me	0.86	0.0	8.4	0.0
3	alkene	Me₂C=CMe₂	1.74	0.84	20.4	12.0
4	benzene	Me–Ph	2.32	1.32	21.4	13.0
5	alkyne	Me–C≡C–R^a	1.86	0.96		
6	nitrile	Me–CN	2.04	1.14	1.8	−6.6
7	acid	Me–CO₂H	2.10	1.20	20.9	11.5
8	ester	Me–CO₂Me	2.08	1.18	20.6	11.2
9	amide	Me–CONHMe	2.00	1.10	22.3	13.9
10	ketone	Me₂C=O	2.20	1.30	30.8	21.4
11	aldehyde	Me–CHO	2.22	1.32	30.9	21.5
12	sulfide	Me₂S	2.13	1.23	18.1	9.7
13	sulfoxide	Me₂S=O	2.71	1.81	41.0	32.6
14	sulfone	Me₂SO₂	3.14	2.24	44.4	36.0
15	amine	Me–NH₂	2.41	1.51	26.9	18.5
16	amide	MeCONH–Me	2.79	1.89	26.3	17.9
17	nitro	Me–NO₂	4.33	3.43	62.5	53.1
18	ammonium salt	Me₄N⁺Cl[−]	3.20	2.10	58.0	49.6
19	alcohol	Me–OH	3.50	2.60	50.3	44.3
20	ether	Me–OBu	3.32	2.42	58.5	50.1
21	enol ether	Me–OPh	3.78	2.88	55.1	46.7
22	ester	Me–CO₂Me	3.78	2.88	51.5	47.1
23	phosphonium salt	Ph₃P⁺–Me	3.22	2.32	11.0	2.2

^aR=CH₂OH; compound is but-2-yn-1-ol.

The effects of groups based on carbon (the methyl group is joined directly to another carbon atom) appear in entries 2 to 11. All the electron-withdrawing groups based on carbonyl and cyanide have about the same effect (1.1–1.3 ppm downfield shift from 0.9 ppm). Groups based on nitrogen (Me–N bond) show a similar progression through amine, ammonium salt, amide, and nitro compound (entries 15–18). Finally, all the oxygen-based groups (Me–O bond) show large shifts (entries 19–22).

Effects of substituents on CH₂ groups

It is more difficult to give a definitive list for CH₂ groups as they have two substituents. In the table below we set one substituent as phenyl (Ph) just because so many compounds of this kind are available, and give the actual shifts relative to PhCH₂CH₃ for protons (2.64 ppm) and PhCH₂CH₃ for carbon (28.9 ppm), again comparing the substituent with the CH skeleton.

If you compare the shifts caused on a CH₂ group by each functional group in the table below with the shifts caused on a CH₃ group by the same functional group in the table on p. 423 you will see that they are broadly the same.

Chemical shifts of CH₂ groups bonded to phenyl and functional groups

	Functional group	Compound	δ_H , ppm	$\delta_H - 2.64$, ppm	δ_C , ppm	$\delta_C - 28.9$, ppm
1	silane	PhCH ₂ -SiMe ₃	?	?	27.5	-1.4
2	hydrogen	PhCH ₂ -H	2.32	-0.32	21.4	-7.5
3	alkane	PhCH ₂ -CH ₃	2.64	0.00	28.9	0.0
4	benzene	PhCH ₂ -Ph	3.95	1.31	41.9	13.0
5	alkene	PhCH ₂ -CH=CH ₂	3.38	0.74	41.2	12.3
6	nitrile	PhCH ₂ -CN	3.70	1.06	23.5	-5.4
7	acid	PhCH ₂ -CO ₂ H	3.71	1.07	41.1	12.2
8	ester	PhCH ₂ -CO ₂ Me	3.73	1.09	41.1	12.2
9	amide	PhCH ₂ -CONEt ₂	3.70	1.06	?	?
10	ketone	(PhCH ₂) ₂ C=O	3.70	1.06	49.1	20.2
11	thiol	PhCH ₂ -SH	3.69	1.05	28.9	0.0
12	sulfide	(PhCH ₂) ₂ S	3.58	0.94	35.5	6.6
13	sulfoxide	(PhCH ₂) ₂ S=O	3.88	1.24	57.2	28.3
14	sulfone	(PhCH ₂) ₂ SO ₂	4.11	1.47	57.9	29.0
15	amine	PhCH ₂ -NH ₂	3.82	1.18	46.5	17.6
16	amide	HCONH-CH ₂ Ph	4.40	1.76	42.0	13.1
17	nitro ^a	PhCH ₂ -NO ₂	5.20	2.56	81.0	52.1
18	ammonium salt	PhCH ₂ -NMe ₃ ⁺	4.5/4.9		55.1	26.2
19	alcohol	PhCH ₂ -OH	4.54	1.80	65.3	36.4
20	ether	(PhCH ₂) ₂ O	4.52	1.78	72.1	43.2
21	enol ether	PhCH ₂ -OAr ^a	5.02	2.38	69.9	41.0
22	ester	MeCO ₂ -CH ₂ Ph	5.10	2.46	68.2	39.3
23	phosphonium salt	Ph ₃ P ⁺ -CH ₂ Ph	5.39	2.75	30.6	1.7
24	chloride	PhCH ₂ -Cl	4.53	1.79	46.2	17.3
25	bromide	PhCH ₂ -Br	4.45	1.81	33.5	4.6

^a Compound is (4-chloromethylphenoxy)methyl)benzene.

Shifts of a CH group

We can do the same with a CH group, and in the left-hand side of the table below we take a series of isopropyl compounds, comparing the measured shifts with those for the central proton (CHMe_2) or carbon (CHMe_2) of 2-methylpropane. We set two of the substituents as methyl groups and just vary the third. Yet again the shifts for the same substituent are broadly the same.

Effects of α and β substitution on ^1H and ^{13}C NMR shifts in Me_2CHX^a

X	Effects on C_α ($\text{Me}_2\text{CH-X}$), ppm				Effects on C_β ($\text{Me}_2\text{CH-X}$), ppm			
	δ_{H}	$\delta_{\text{H}} - 1.68$	δ_{C}	$\delta_{\text{C}} - 25.0$	δ_{H}	$\delta_{\text{H}} - 0.9$	δ_{C}	$\delta_{\text{C}} - 8.4$
Li			10.2	-14.8			23.7	17.3
H	1.33	-0.35	15.9	-9.1	0.91	0.0	16.3	7.9
Me	1.68	0.00	25.0	0.0	0.89	0.0	24.6	16.2
$\text{CH}=\text{CH}_2$	2.28	0.60	32.0	7.0	0.99	0.09	22.0	13.6
Ph	2.90	1.22	34.1	9.1	1.24	0.34	24.0	15.6
CHO	2.42	0.74	41.0	16.0	1.12	0.22	15.5	7.1
COMe	2.58	0.90	41.7	16.7	1.11	0.21	27.4	19.0
CO_2H	2.58	0.90	34.0	4.0	1.20	0.30	18.8	10.4
CO_2Me	2.55	0.87	33.9	8.9	1.18	0.28	19.1	10.7
CONH_2	2.40	0.72	34.0	9.0	1.08	0.18	19.5	11.1
CN	2.71	1.03	20.0	-5.0	1.33	0.43	19.8	11.4
NH_2	3.11	1.43	42.8	17.8	1.08	0.18	26.2	17.8
NO_2	4.68	3.00	78.7	53.7	1.56	0.66	20.8	12.4
SH	3.13	1.45	30.6	5.6	1.33	0.43	27.6	19.2
<i>S</i> -Pr	3.00	1.32	33.5	8.5	1.27	0.37	23.7	15.3
OH	4.01	2.33	64.2	39.2	1.20	0.30	25.3	16.9
<i>O</i> -Pr	3.65	1.97	68.4	43.4	0.22	0.22	22.9	14.5
O_2CMe	5.00	3.32	67.6	42.6	1.22	0.32	21.4(8)	17.0(4)
Cl	4.19	2.51	53.9	28.9	1.52	0.62	27.3	18.9
Br	4.29	2.61	45.4	20.4	1.71	0.81	28.5	20.1
I	4.32	2.36	31.2	6.2	1.90	1.00	21.4	13.0

^a There is coupling between the CH and the Me_2 groups in the proton NMR.

Shifts in proton NMR are easier to calculate and more informative than those in carbon NMR

This final table, on p. 426, helps to explain something we have avoided so far. Correlations of shifts caused by substituents in proton NMR really work very well. Those in ^{13}C NMR work much less well and more complicated equations are needed. More strikingly, the proton shifts often seem to fit better with our understanding of the chemistry of the compounds. There are two main reasons for this.

First, the carbon atom is much closer to the substituent than the proton. In the compounds in the table on p. 423 the methyl carbon atom is directly bonded to the substituent, while the protons are separated from it by the carbon atom of the methyl group. If the functional group is based on a large electron-withdrawing atom like sulfur, the protons will experience a simple inductive electron withdrawal and have a proportional downfield shift. The carbon atom is close enough to the sulfur atom to be shielded as well by the lone-pair electrons in the large 3sp^3 orbitals. The proton shift caused by S in Me_2S is about the same (1.23 ppm) as that caused by a set of more or less equally strong electron-withdrawing groups like CN (1.14 ppm) or ester (1.18 ppm). The carbon shift (9.7 ppm) is less than that caused by an ester (11.2 ppm) but much *more* than that caused by CN, which actually shifts the carbon upfield (-6.6 ppm) relative to the effect of a methyl group.

Approximate additive functional group (X) shifts in ^1H NMR spectra

Entry	Functional group X	^1H NMR shift difference ^a , ppm
1	alkene ($-\text{C}=\text{C}$)	1.0
2	alkyne ($-\text{C}\equiv\text{C}$)	1.0
3	phenyl ($-\text{Ph}$)	1.3
4	nitrile ($-\text{C}\equiv\text{N}$)	1.0
5	aldehyde ($-\text{CHO}$)	1.0
6	ketone ($-\text{COR}$)	1.0
7	acid ($-\text{CO}_2\text{H}$)	1.0
8	ester ($-\text{CO}_2\text{R}$)	1.0
9	amide ($-\text{CONH}_2$)	1.0
10	amine ($-\text{NH}_2$)	1.5
11	amide ($-\text{NHCOR}$)	2.0
12	nitro ($-\text{NO}_2$)	3.0
13	thiol ($-\text{SH}$)	1.0
14	sulfide ($-\text{SR}$)	1.0
15	sulfoxide ($-\text{SOR}$)	1.5
16	sulfone ($-\text{SO}_2\text{R}$)	2.0
17	alcohol ($-\text{OH}$)	2.0
18	ether ($-\text{OR}$)	2.0
19	aryl ether ($-\text{OAr}$)	2.5
20	ester ($-\text{O}_2\text{CR}$)	3.0
21	fluoride ($-\text{F}$)	3.0
22	chloride ($-\text{Cl}$)	2.0
23	bromide ($-\text{Br}$)	2.0
24	iodide ($-\text{I}$)	2.0

^a To be added to 0.9 ppm for MeX , 1.3 ppm for CH_2X , or 1.7 ppm for CHX .

Second, the carbon shift is strongly affected not only by what is directly joined to that atom (α position), but also by what comes next (β position). The right-hand half of the table on p. 424 shows what happens to methyl shifts when substituents are placed on the next carbon atom. There is very little effect on the proton spectrum: all the values are much less than the shifts caused by the same substituent on a methyl group in the table on p. 423. Carbonyls give a downfield shift of about 1.2 ppm when directly joined to a methyl group, but only of about 0.2 ppm when one atom further away. By contrast, the shifts in the carbon spectrum are of the same order of magnitude in the two tables, and the β shift may even be greater than the α shift! The CN group shifts a directly bonded methyl group upfield (-6.6 ppm) when directly bonded, but downfield (14.4 ppm) when one atom further away. This is an exaggerated example, but the point is that these carbon shifts must *not* be used to suggest that the CN group is electron-donating in the α position and electron-withdrawing in the β position. The carbon shifts are erratic but the proton shifts give us useful information and are worth understanding as a guide to both structure determination and the chemistry of the compound.

When you use this table and are trying to interpret, say, a methyl group at 4.0 ppm then you have no problem. Only one group is attached to a methyl group so you need a single shift value—it might be a methyl ester, for example. But when you have a CH_2 group at 4.5 ppm and you are interpreting a downfield shift of 3.2 ppm you must beware. There are *two* groups attached to each CH_2 group and you might need a single shift of about 3 ppm (say, an ester again) or two shifts of 1.5 ppm, and so on. The shifts are additive.

Further reading

A reminder: you will find it an advantage to have one of the short books on spectroscopic analysis to hand as they give explanations, comprehensive tables of data, and problems. We recommend *Spectroscopic Methods in Organic Chemistry* by D. H. Williams and Ian Fleming, McGraw-Hill, London, 6th edn, 2007.

Other books include R. M. Silverstein, F. X. Webster, and D. J. Kiemle, *Spectrometric Identification of Organic Compounds*,

Wiley, 2005 and a book of problems: L. D. Field, S. Sternhell, and J. R. Kalman, *Organic Structures from Spectra*, 3rd edn, Wiley, 2003.

The ^{13}C NMR of ketene was reported by J. Firl and W. Runger, *Angew. Chem. Int. Ed.*, 1973, **12**, 668, the tetrahedrane/cyclobutadiene story is expounded by G. Maier in *Angew. Chem. Int. Ed.*, 1988, **27**, 309, and the *Lycorea* sex pheromone story by G. Meinwald and team, *Science*, 1968, **164**, 1174.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>